

Frontend – Assignment 1

Q1. HTTP Request–Response Cycle

Answer: The HTTP Request–Response Cycle is the process through which a client, usually a web browser, communicates with a web server to request and receive a resource. The major steps are:

1. **Client request:** The user requests a webpage or resource through the browser.
2. **HTTP request:** The browser sends an HTTP request to the web server.
3. **Server processing:** The server receives and processes the request.
4. **HTTP response:** The server sends a response containing the requested resource or an appropriate result.
5. **Browser displays the result:** The browser receives the response and displays or updates the webpage.

Diagram:

Client / Browser → HTTP Request → Web Server
Client / Browser ← HTTP Response ← Web Server

Q2. Introduction to Web Technologies

Answer:

1. **HTML:** HTML (HyperText Markup Language) is used to create the structure and content of a webpage. It defines elements such as headings, paragraphs, images, links, tables and forms.
2. **CSS:** CSS (Cascading Style Sheets) is used to style and design HTML elements. It controls properties such as colors, fonts, spacing, borders and layout.
3. **JavaScript:** JavaScript is a programming language used to make webpages dynamic and interactive. It can respond to user actions, modify webpage content and perform operations in the browser.
4. **Bootstrap:** Bootstrap is a frontend framework that provides ready-made CSS classes, responsive layouts and UI components. It helps developers build responsive webpages more quickly.
5. **jQuery:** jQuery is a JavaScript library that simplifies common tasks such as DOM manipulation, event handling and animations.
6. **React.js:** React.js is a JavaScript library used for building user interfaces using reusable components. It is commonly used for interactive and component-based frontend applications.

Q3. Client-Side vs. Server-Side Technologies

Answer: Client-side technologies mainly execute in the user's browser, while server-side technologies execute on the web server.

Basis	Client-Side	Server-Side
Execution	Browser	Web server
Purpose	User interface and interaction	Processing, business logic and data operations
Examples	HTML, CSS, JavaScript	Node.js, PHP, Python, Java

Basis	Client-Side	Server-Side
Typical role	Displays and updates the webpage	Processes requests and sends responses

For example, JavaScript may handle an interaction in the browser, while a server-side program can process a request, communicate with a database and return a response.

Q4. Block-Level Elements in HTML

Answer: Block-level elements are HTML elements that generally start on a new line and occupy the available horizontal space of their containing block.

Characteristics:

- Generally begin on a new line.
- Usually occupy the available width.
- Can be used to organize the structure of a webpage.
- Their exact layout can also be changed using CSS.

Examples: <div>, <p>, <h1>, <section>, <header>.

Q5. Semantic HTML Elements

Answer: Semantic HTML elements are elements whose names clearly describe the meaning or role of the content they contain. They make the structure of a webpage easier to understand for developers, browsers and assistive technologies.

Importance:

- Improves code readability and structure.
- Helps communicate the purpose of different sections of a webpage.
- Supports accessibility.
- Provides meaningful document structure.

Examples: <header>, <nav>, <main>, <section>, <article>, <aside>, <footer>.

Q6. Creating a Table Using HTML

Answer: The following HTML code creates the required table:

```
<table border="1">
<tr>
<th>Sr. No.</th>
<th>Name</th>
<th>Designation</th>
<th>Department</th>
</tr>
<tr>
<td>1</td>
<td>John</td>
<td>Manager</td>
<td>Sales</td>
</tr>
<tr>
<td>2</td>
<td>Smith</td>
<td>Sr. Manager</td>
<td>Marketing</td>
</tr>
<tr>
<td>3</td>
```

```

<td>Jane</td>
<td>Manager</td>
<td>HR</td>
</tr>
</table>

```

Q7. Creating a Nested List Using HTML

Answer: An unordered list can be nested inside the main list item to create the required structure.

```

<ul>
<li>River
<ul>
<li>Ganga</li>
<li>Yamuna</li>
<li>Brahmaputra</li>
<li>Narmada</li>
</ul>
</li>
</ul>

```

Q8. Types of CSS

Answer: There are three common ways to apply CSS to an HTML document.

1. Inline CSS: CSS is written directly inside the HTML element using the style attribute.

```

<p style="color: red;">Hello</p>

```

2. Internal CSS: CSS is written inside a <style> element, usually inside the <head> section.

```

<style>
p {
color: blue;
}
</style>

```

3. External CSS: CSS is written in a separate .css file and linked to the HTML document.

```

<link rel="stylesheet" href="style.css">

```

Q9. Practical Understanding

Answer: HTML, CSS and JavaScript work together, but each has a different role:

Technology	Role
HTML	Provides the structure and content of the webpage.
CSS	Controls the appearance, styling and layout of the webpage.
JavaScript	Adds dynamic behavior, interaction and functionality.

In simple terms: **HTML = structure, CSS = presentation, JavaScript = behavior.**

Q10. Short Answer

a. What is the role of Bootstrap in web development?

Bootstrap provides a collection of ready-made CSS classes, responsive layout utilities and UI components. It helps developers create responsive and consistent interfaces more quickly.

b. What is jQuery and why was it widely used in web development?

jQuery is a JavaScript library that simplifies common tasks such as DOM manipulation, event handling and animations. It became widely used because it made many browser-side JavaScript tasks shorter and easier to write, especially when browser differences were more significant.

c. What is React.js and what problem does it solve in frontend development?

React.js is a JavaScript library for building user interfaces with reusable components. It helps developers organize complex interfaces into smaller components and efficiently manage changes in the UI.

End of Assignment 1